

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
Student Name:	TETASWI REDDY · K	Reg. No.:	192371036
Section / Batch:	B.Tech CS-BIOSCI	Date:	30/07/2026

Instructions to Students

- Answer ALL questions. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type		Questions	Marks
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1 Q2, Q3	40 60 (2X30)
GRAND TOTAL:			100 Marks

SECTION A – APPLICATION BASED QUESTIONS

LINEAR SEARCH

Introduction -

→ Linear search is also called sequential search is one of the simplest searching algorithms used to locate a specific element in list or array.

→ It works by checking each element one after another from the beginning of the list until the required element is found or until entire list has been searched.

→ Unlike Binary search, linear search does not require the data to be sorted.

* It is widely used for searching small datasets or unsorted collections.

Defⁿ - Linear Searching technique in which every element of the array is compared with the target element until a match is found or all elements have been examined.

Characteristics of Linear Search -

Searches element one by one from left to right.
works with sorted and unsorted datasets.

Very easy to understand and implement

Requires no preprocessing or sorting.

May become inefficient for large datasets because it

can require checking every element.

For a dataset containing n elements:

Best Case: 1 Comparison

Average Case: $(n+1)/2$ comparisons

Worst Case: n comparisons.

Advantages of Linear Search -

- * Simple and easy to implement.
- * No preprocessing or sorting required.
- * Suitable for small datasets.

Disadvantages of Linear Search -

- * Slow for large datasets
- * may require checking every element.
- * Less efficient than Binary search on sorted data.
- * Execution time increases linearly with size of dataset.

Working Principle -

- * Start the search from first element.
- * Compare the current element with target value.
- * If values are equal, stop the search.
- * If values are not equal, move to next element
- * Repeat the process until value is found (or) Array ends.

LINEAR SEARCH - Element at Last Position

Given,

$A = [41, 26, 13, 57, 88, 34, 72]$

Key = 72.

Step	Element Compared	Comparison	Result
1	41	$41 == 72$	No
2	26	$26 == 72$	No
3	13	$13 == 72$	No
4	57	$57 == 72$	No
5	88	$88 == 72$	No
6	34	$34 == 72$	No
7	72	$72 == 72$	Found.

It takes 7 comparisons to find the element.

Index Found: 6

Time Complexity Analysis

$\therefore$ Element is found at last position, it is Worst Case.

Worst Case: $O(n)$

If element is found in the first position, it is Best Case.

Best Case: $O(1)$

Even if element is in the Middle, then it is Average

Case:

Average Case: $O(n)$

$\therefore$ This Problem comes under Worst Case.

Verification of Correctness -

This algorithm examines each element one by one:

41 $\rightarrow$ 26 $\rightarrow$ 13 $\rightarrow$ 57 $\rightarrow$ 88 $\rightarrow$ 34 $\rightarrow$ 72.

When element is found, the search stops.

Algorithm -

For $i \leftarrow 0$ to $n-1$ do

if $A[i] == \text{key}$ then

return i

end if

end for

return "not found"

Efficiency Interpretation -

When target element appears at last position, Linear Search must inspect every preceding element before finding it.

This results in maximum possible No. of comparisons, making search least efficient for this dataset.

Although linear search is simple and works well for small or unsorted datasets, its performance decreases as the dataset size increases because the search time grows linearly with no. of elements.

Code of Conduct

I Tejani Reddy K (192371036) (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Tejani Reddy K

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (40 Marks)	4	4	4	4	4	40
Q2 (30 Marks)	4	4	4	4	4	30
Q3 (30 Marks)	4	3	3	2	2	22
Overall Total (100)						92

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____